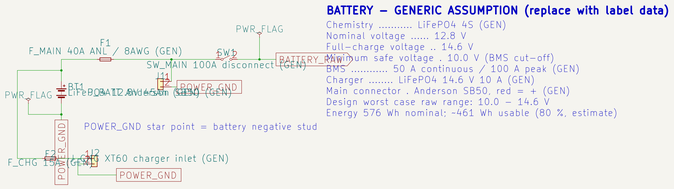
Battery and Charging

KIRO Wiring V2 | Battery_and_Charging

Battery, main fuse, disconnect, charger inlet and POWER_GND star point



BATTERY — GENERIC ASSUMPTION (replace with label data)

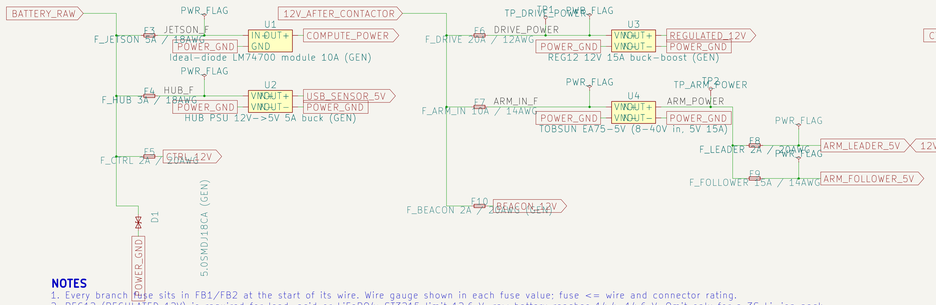
Chemistry ..... LiFePO4 4S (GEN)
Nominal voltage ..... 12.8 V
Full-charge voltage .. 14.6 V
Minimum safe voltage . 10.0 V (BMS cut-off)
BMS ..... 50 A continuous / 100 A peak (GEN)
Charger ..... LiFePO4 14.6 V 10 A (GEN)
Main connector . Anderson SB50, red = + (GEN)
Design worst case raw range: 10.0 – 14.6 V
Energy 576 Wh nominal: ~461 Wh usable (80 %, estimate)

J2/F2: charger joins on the battery side of SW_MAIN, so the robot can charge while switched off.
Reverse polarity: J_BATT must be keyed; compute branch adds an ideal-diode stage (Power_Distribution).
F_MAIN within 150 mm of BT1+. 40 A: above 25.5 A peak load, below 50 A BMS and 8 AWG rating.

Power Distribution

KIRO Wiring V2 | Power_Distribution

FB1 = unswitched fuse box (compute side) | FB2 = switched fuse box after contactor K1 (actuator side)



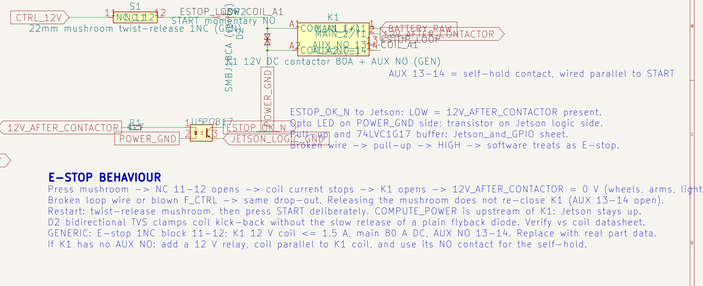
NOTES

1. Every branch fuse sits in FB1/FB2 at the start of its wire. Wire gauge shown in each fuse value: fuse <= wire and connector rating.
2. REG12 (REGULATED_12V) is required for lead-acid or LiFePO4; ST3215 limit 12.6 V, raw battery reaches 14.4–14.6 V. Omit only for a 3S Li-ion pack.
3. U1 reverse-polarity stage protects the Jetson branch. High-current branches rely on keyed J_BATT + F_MAIN.
4. TVS_BUS 18 V stand-off: above LiFePO4 14.6 V charge voltage (GEN — recheck if chemistry differs).
5. Pre-charge not fitted: generic REG12 input capacitance is small for an 80 A contactor. Re-check with the real REG12.
6. Arm 5 V rail: EA75-5V as in V1. Follower servo type on V2 (7.4 V or 12 V) is TBD — 5 V is safe for both.
7. HUB PSU stays on the unswitched side so cameras, LiDAR and adapter logic keep working after an E-stop.
8. #FLG = ERC power flags only (rail enters through a passive fuse). Not physical parts.

Safety EStop and Contactor

KIRO Wiring V2 | Safety_EStop_and_Contactor

Hard-wired self-holding contactor loop. No software in the safety path.



E-STOP BEHAVIOUR

Press mushroom -> NC 11–12 opens -> coil current stops -> K1 opens -> 12V_AFTER_CONTACTOR = 0 V (wheels, arms, light).
Broken loop wire or blown F_CTRL -> same drop-out. Releasing the mushroom does not re-close K1 (AUX 13–14 open).
Restart: twist-release mushroom, then press START deliberately. COMPUTE_POWER is upstream of K1: Jetson stays up.
D2 bidirectional TVS clamps coil kick-back without the slow release of a plain flyback diode. Verify vs coil datasheet.
GENERIC: E-stop 1NC block 11–12; K1 12 V coil <= 1.5 A, main 80 A DC, AUX NO 13–14. Replace with real part data.
If K1 has no AUX NO: add a 12 V relay, coil parallel to K1 coil, and use its NO contact for the self-hold.

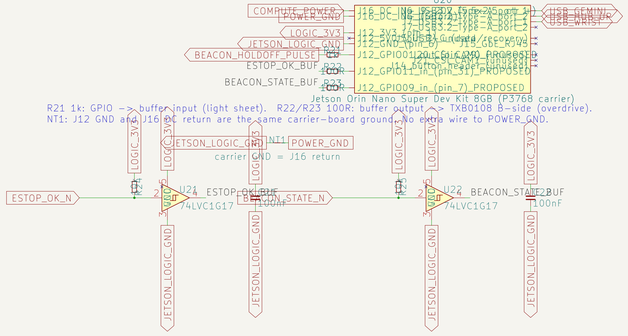
Jetson and GPIO

KIRO Wiring V2 | Jetson_and_GPIO

NVIDIA Jetson Orin Nano Super Developer Kit, reference carrier: J16 DC jack, J12 40-pin header, J6/J7 USB

J12 HEADER FACTS (NVIDIA carrier spec)

All J12 signals 3.3 V. Non-I2C pins go through TI TXB0108 level translators: drive only +/-20 uA.
External pull-ups/pull-downs on these pins must be > 50 kOhm. Never load a J12 pin directly.
Therefore every GPIO here meets a 74LVC1G17 buffer (CMOS input, push-pull output).
Pins 29/31/7 are PROPOSED. Pinmux, boot state and state after cleanup: bench-verify (TBD).
J16: 9–20 V, 5.5 x 2.5 mm, centre positive, max 3.5 A. Raw battery 10–14.6 V is inside range.
Do NOT use hardware PWM for BEACON_HOLDOFF_PULSE: it must stop when software freezes.



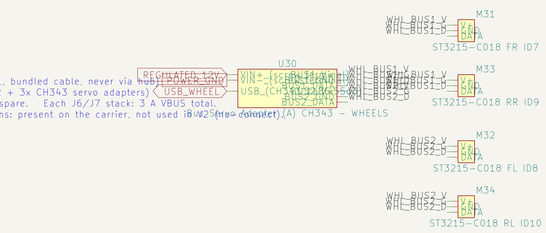
USB PORT ASSIGNMENT

J6 port 1: Orbbec Gemini 335 (USB 3.2 Gen1, bundled cable, never via hub) | J7 port 1: Suyin wrist webcam.
J6 port 2: powered USB hub upstream (LiDAR + 3x CH343 servo adapters) | J7 port 2: spare.
J7 port 1: Suyin wrist webcam. J7 port 2: spare. Each J6/J7 stack: 3 A VBUS total.
J15 GbE, J5 USB-C, J20/J21 CSI, J14 buttons: present on the carrier, not used in this project.

Wheel Drive

KIRO Wiring V2 | Wheel_Drive

4 x Feetech/Waveshare ST3215-C018 smart serial servos on one CH343 bus adapter (no generic motor driver)

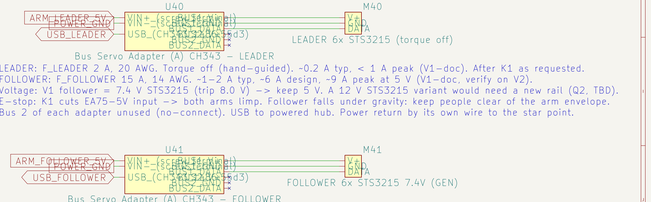


Chain 1 (BUS1): adapter -> FR ID7 -> RR ID9. Chain 2 (BUS2): adapter -> FL ID8 -> RL ID10. Max two servos per Molex 5264 contact (~3 A).
Supply: REGULATED_12V (ST3215 6–12.6 V: V1 Max_Voltage_Limit 14.0 V). Adapter powered via screw terminal, not its DC jack.
Current: idle ~1.2 A total, design 6 A, all four stalled 10.8 A (2.7 A each). Feed 14 AWG to screw terminal; 20 AWG servo leads.
Servo cable: 3-pin Molex 5264 (V+, GND, DATA). Physical pin order: verify against Waveshare ST3215 wiki before crimping (TBD).
Data: half-duplex TTL 1 Mbps. USB_WHEEL via powered hub. Servo GND returns to the adapter, never through the USB cable shield.
E-stop: K1 removes REGULATED_12V -> servos unpowered, no holding torque, robot coasts. Speed feedback over bus = ROBOT_MOVING.

Leader Follower Arms

KIRO Wiring V2 | Leader_Follower_Arms

LeRobot leader + follower arms, each on its own CH343 bus adapter and its own fused 5 V branch



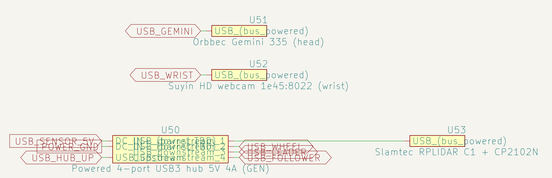
LEADER: F_LEADER 2 A, 20 AWG, Torque off (hand-guided). ~0.2 A typ, < 1 A peak (V1-doc). After K1 as requested.
FOLLOWER: F_FOLLOWER 15 A, 14 AWG. ~1–2 A typ, ~6 A design, ~9 A peak at 5 V (V1-doc, verify on V2).
Voltage: V1 follower = 7.4 V STS3215 (trip 8.0 V) -> keep 5 V. A 12 V STS3215 variant would need a new rail (Q2, TBD).
E-stop: K1 cuts EA75-5V input -> both arms limp. Follower falls under gravity: keep people clear of the arm envelope.
Bus 2 of each adapter unused (no-connect). USB to powered hub. Power return by its own wire to the star point.

Adapter feed via screw terminal. Vendor nominal input 9–12.6 V: V1 runs these adapters from 5 V successfully (V1-measured) — ke

USB Hubs Cameras and LiDAR

KIRO Wiring V2 | USB_Hubs_Cameras_and_LiDAR

Only the installed hub, two cameras and one LiDAR. No other sensors.



DEVICE TABLE

Gemini 335 | USB bus 5 V | 896 mA max, <6.5 W | USB 3.2 Gen1 | bundled cable | Jetson J6 port 1 direct
Wrist webcam | USB bus 5 V | needs 500 mA | USB 2.0 | camera cable | Jetson J7 port 1 direct
RPLIDAR C1 | 5 V via USB | 230 mA typ, peak TBD | UART 460800 inside, USB outside (CP2102N) | hub DS1
USB hub | DC in 5 V 4 A, barrel 5.5 x 2.1 mm, centre positive (GEN — check against real hub label)
Hub barrel: if the real hub is 3.5x1.35 or 5.5x2.5 mm, change the plug; never trust '5 mm' alone.
Shielding: use shielded USB cables; route away from servo and 12 V harnesses. No motor current returns via USB.
LiDAR start-up (V1): plug sensor cable into adapter with USB unplugged, then connect USB. Motor spin-up after host opens port.

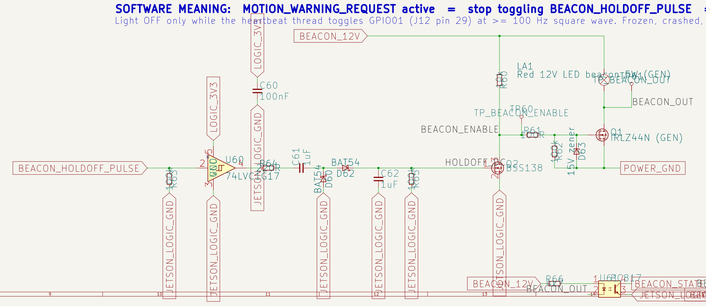
Red Motion Warning Light

KIRO Wiring V2 | Red_Motion_Warning_Light

Fail-to-warning: light is ON whenever K1 is closed, unless the Jetson actively holds it OFF with a live pulse train

SOFTWARE MEANING: MOTION_WARNING_REQUEST active = stop toggling BEACON_HOLDOFF_PULSE = light ON

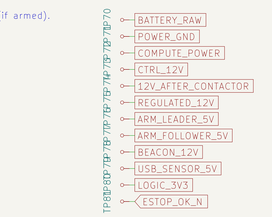
Light OFF only while the heartbeat thread toggles GPIO01 (J12 pin 29) at >= 100 Hz square wave. Frozen, crashed, booting or powered-off Jetson = light ON (if armed).



Connectors and Test Points

KIRO Wiring V2 | Connectors_and_Test_Points

Test-point panel on every cross-sheet rail, plus the connector schedule



CONNECTOR SCHEDULE (rating must exceed fuse; pinout verified before crim

J_BATT (J1) Anderson SB50, keyed, 50 A ..... GEN
J_CHG (J2) XT60 charger inlet ..... GEN
J16 Jetson 5.5 x 2.5 mm barrel, centre +, 3.5 A max , NVIDIA spec
Adapter VIN screw terminal (not the 5.5 x 2.1 jack) .. Waveshare
Servo bus Molex 5264 3-pin, use factory servo leads
Hub DC in 5.5 x 2.1 mm, centre +, 5 V 4 A ..... GEN
J_BEACON (J60) 2-pin 5 A, pin 1 = lamp + ... GEN
ESTOP / START panel wiring, 20 AWG, ferrules ..... GEN
K1 main M6 ring lugs, 80 A DC contact ..... GEN
Signal harness J12 pins 1/6/29/31/7 to buffer board, 2.54 mm Dupont, strain-relieved

TEST-POINT USE

Measure all rails against TP_POWER_GND at the star point. Local TPs:
TP1 DRIVE_POWER, TP2 ARM_POWER (Power_Distribution): TP60 BEACON_ENABLE,
TP61 BEACON_OUT (Red_Motion_Warning_Light). Strain relief on every harness exit.

Flattened from hierarchical sheets for single-file delivery
Genie Enrtech
Sheets: /
File: KIRO_Wiring_v2_Xlcad.sch
Title: KIRO Wiring V2 (Flat)
Size: A1 Date: 2026-11-12 Rev: Rst-1
KiCad 8.0.4 10.5.7 Id: 1/1